

Guardrails for AI Agents

Making AI agents safe and reliable

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WORKSHOP FLOW

What we build

- Data Engineering Zoomcamp FAQ agent
- One search tool over the FAQ JSON
- Input and output guardrails
- Concurrent checks for lower latency

3 hour workshop with exercises, Q&A, and breaks

SEGMENT 1

Introduction to agents

- What AI agents are and how they work
- Agents vs standalone LLM calls
- Tool calling as the agent capability layer
- When a FAQ bot needs an agent loop
- Brief overview of agent frameworks

25 minutes

Q&A

5 minutes

SEGMENT 2

Build the FAQ agent

- Set up the environment and dependencies
- Load the Data Engineering Zoomcamp FAQ JSON
- Create a FAQ search tool
- Build the agent with instructions
- Run the agent and inspect failures

30 minutes

Exercise

Q&A

5 minutes

Break

5 minutes

SEGMENT 3

Input guardrails

- What input guardrails are and when to use them
- Guardrail decisions with Pydantic
- Topic guardrail for FAQ boundaries
- Attach the guardrail to the agent
- Handle blocked questions

35 minutes

Exercise

Q&A

5 minutes

SEGMENT 4

Output guardrails

- What output guardrails are and when to use them
- Safety checks for agent responses
- Guardrail for unsafe promises
- Use input and output checks together
- Test deadline, refund, and policy edge cases

35 minutes

Exercise

Q&A

5 minutes

Break

5 minutes

SEGMENT 5

Multiple guardrails

- Chain multiple guardrails together
- Add academic integrity checks
- Stream responses with guardrails
- Run checks concurrently
- Keep the pattern framework-independent

30 minutes

Exercise

Q&A

5 minutes

WRAP-UP

Next steps

- Summary of key takeaways
- When to use each guardrail type
- Further reading and resources
- Final Q&A

15 minutes